

Requirements Table

Lab 1 — Requirements Engineering & UML Use-Case Modelling

Problem Statement #53 — Freelance Content Creator Escrow Platform

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1. Scenario Recap

A freelance contract management system that lets **Content Creators** and **Client Sponsors** agree on deliverable milestones, fund those milestones into escrow, exchange watermarked draft assets for review, and release locked payments only after an explicit sponsor sign-off.

Actors identified: Content Creator, Client Sponsor, Payment Gateway (external system), Platform Admin.

2. Functional Requirements (FR-001 – FR-005)

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale (short)
FR-001	Functional	The system shall allow a Content Creator to submit a draft media deliverable against a milestone for Client Sponsor review, and shall release the locked milestone payment to the creator's balance only after the sponsor records an approval sign-off.	High	Pass: milestone state moves SUBMITTED → APPROVED → PAID and the escrowed amount is credited to the creator balance within 1 business day of sign-off. Fail: funds move to the creator while the milestone sign-off flag is <code>false</code> .	Core escrow guarantee — it is the single behaviour that makes the platform trustworthy to both parties.
FR-002	Functional	The system shall allow a Client Sponsor to create a contract by defining 1–20 milestones, each with a title, due date and amount, and shall require the full contract value to be funded into escrow before the contract becomes ACTIVE.	High	Pass: a contract with an unfunded balance > ₹0 stays in DRAFT and the creator cannot upload against it. Fail: contract reaches ACTIVE while escrow balance < total milestone value.	Prevents the classic freelance failure mode of a client disappearing without funds committed.
FR-003	Functional	The system shall serve every draft deliverable to the Client Sponsor only as an automatically watermarked preview (creator name + contract ID + timestamp burned in), and shall block download of the original asset until the milestone is marked PAID.	High	Pass: a GET on the original-asset URL before payment returns HTTP 403, and the preview returned carries a visible watermark. Fail: an un-watermarked original is retrievable pre-payment.	Protects the creator's unpaid work from being used without release of the escrowed funds.
FR-004	Functional	The system shall allow a Client Sponsor to reject a submitted draft with a mandatory written reason, which returns the milestone to REVISION_REQUESTED and lets the Content Creator upload a new version while retaining all previous versions.	Medium	Pass: rejection without a reason of ≥ 20 characters is refused; after a valid rejection the milestone shows REVISION_REQUESTED and both v1 and v2 remain listed and openable. Fail: a prior version is overwritten or lost.	Review cycles are normal in creative work; version history is the evidence trail if a dispute follows.

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale (short)
FR-005	Functional	The system shall allow either party to raise a dispute on a milestone, which immediately freezes that milestone's escrowed funds and notifies a Platform Admin for resolution.	Medium	Pass: within 5 seconds of a dispute being raised the milestone shows <code>DISPUTED</code> , any release attempt is refused, and an admin notification record exists. Fail: a release succeeds on a milestone in <code>DISPUTED</code> state.	Escrow is only meaningful if there is a defined path for the case where the two parties disagree.

3. Non-Functional Requirements (NFR-001 – NFR-002)

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale (short)
NFR-001	Non-functional — <i>Performance & Security</i>	The system shall complete automated watermarking of an uploaded image or video draft (up to 200 MB) and make the preview available in under 5 seconds.	High	Pass: in a load benchmark of 100 concurrent uploads, the 95th-percentile time from upload completion to preview-ready is < 5 s. Fail: p95 ≥ 5 s, or any preview is published without a watermark.	A slow review loop is the main reason sponsors bypass the platform and settle privately, which defeats the escrow.
NFR-002	Non-functional — <i>Security & Auditability</i>	The system shall encrypt all payment and escrow data in transit (TLS 1.3) and at rest (AES-256), and shall write every escrow state change to an append-only audit log that no user role can edit or delete.	High	Pass: a security scan reports no endpoint below TLS 1.3, and an attempted UPDATE / DELETE on an audit-log row by any application role is rejected. Fail: any escrow state change occurs with no corresponding audit entry.	Money movement between strangers must be provable after the fact; the audit log is the evidence used to settle disputes.

4. Requirement → Use-Case Traceability

Req ID	Realised by
FR-001	UC-02 Upload Draft Deliverable, UC-03 Review Draft & Sign Off, UC-04 Release Milestone Payment
FR-002	UC-01 Create Contract & Fund Escrow
FR-003	UC-06 Apply Watermark («include» of UC-02)
FR-004	UC-07 Request Revision («extend» of UC-03)
FR-005	UC-05 Raise Dispute
NFR-001	Constrains UC-06 Apply Watermark
NFR-002	Constrains UC-04 Release Milestone Payment, UC-08 Authorize Escrow Transfer